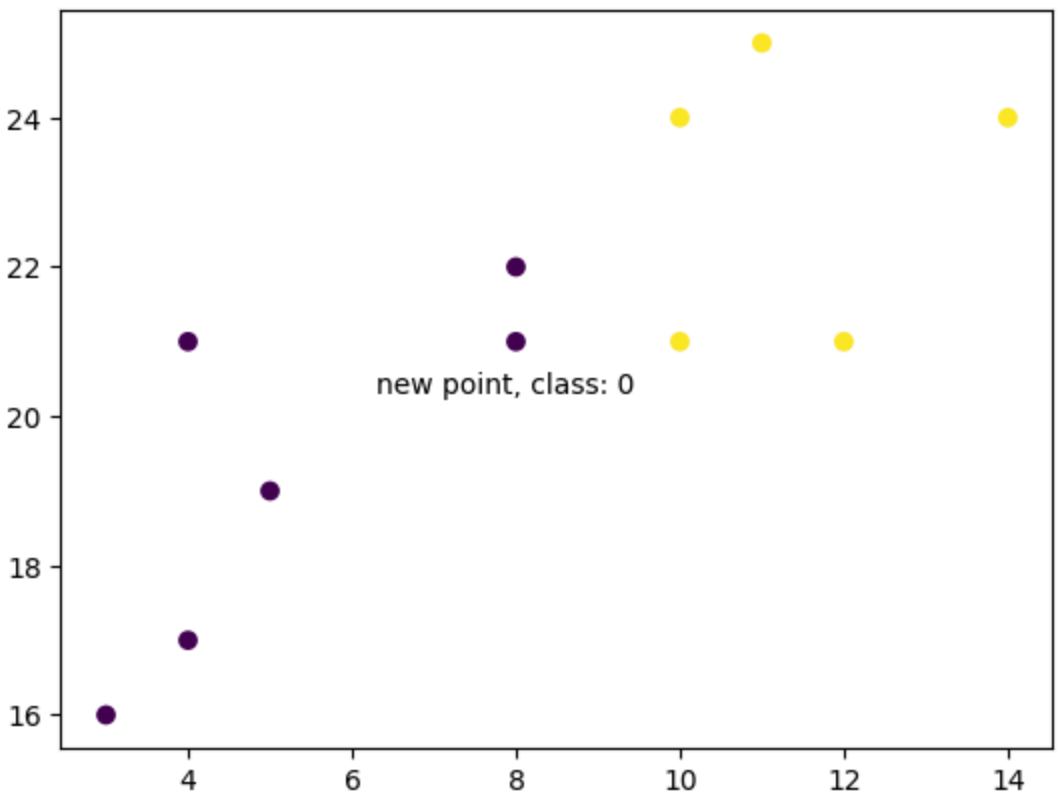
python-ML-Aishwarya / _K_NEAREST_NEIGHBOURS_2(9C).ipynb

Aishwarya-eee Add files via upload 6b9c00a · 2 weeks ago

61 lines (61 loc) · 16.5 KB

Preview Code Blame Raw

In [ ]:

```
import matplotlib.pyplot as plt
from sklearn.neighbors import KNeighborsClassifier
x = [4, 5, 10, 4, 3, 11, 14, 8, 10, 12]
y = [21, 19, 24, 17, 16, 25, 24, 22, 21, 21]
classes = [0, 0, 1, 0, 0, 1, 1, 0, 1, 1]
data = list(zip(x, y))
knn = KNeighborsClassifier(n_neighbors=1)
knn.fit(data, classes)
new_x = 8
new_y = 21
new_point = [(new_x, new_y)]
prediction = knn.predict(new_point)
plt.scatter(x + [new_x], y + [new_y], c=classes + [prediction[0]])
plt.text(new_x - 1.7, new_y - 0.7,
         f"new point, class: {prediction[0]}")
plt.show()
```